

```

tcp46 0 0 *.3306 *.* LISTEN
tcp46 0 0 *.33060 *.* LISTEN
tcp46 0 0 *.59732 *.* LISTEN
tcp6 0 0 2806:268:3403:87.55582 2620:149:a32:f00.443 ESTABLISHED
tcp6 0 0 2806:268:3403:87.55581 2620:149:a41:580.443 TIME_WAIT
tcp4 0 0 192.168.1.36.64506 54.71.177.201.443 ESTABLISHED
tcp6 0 0 2806:268:3403:87.55579 2620:149:a32:f00.443 ESTABLISHED
tcp6 0 0 2806:268:3403:87.55578 2620:149:a32:f00.443 TIME_WAIT
tcp4 0 0 192.168.1.36.64505 17.171.47.23.443 ESTABLISHED
tcp4 0 0 192.168.1.36.64504 17.171.47.23.443 ESTABLISHED
tcp6 0 0 2806:268:3403:87.55575 2620:149:a32:f00.443 ESTABLISHED
tcp6 0 0 2806:268:3403:87.55571 2620:149:a32:f00.443 ESTABLISHED
tcp4 0 0 192.168.1.36.64498 17.145.8.1.443 ESTABLISHED
tcp6 0 0 2806:268:3403:87.55559 2620:149:a32:f00.443 ESTABLISHED
tcp4 0 0 192.168.1.36.64495 20.50.80.213.443 ESTABLISHED
tcp4 0 0 192.168.1.36.64494 23.64.153.219.443 ESTABLISHED
tcp4 0 0 192.168.1.36.64463 17.57.144.27.5223 ESTABLISHED
tcp6 0 0 2806:268:3403:87.55435 2a03:2880:f235:1.5222 ESTABLISHED
tcp6 0 0 2806:268:3403:87.55434 2600:1901:1:3b2:.443 ESTABLISHED

```

nadxiellyvazquez@192 ~ % netstat -a -n -p TCP

(Active Internet connections (including servers))

Proto	Recv-Q	Send-Q	Local Address	Foreign Address	(state)
tcp4	0	0	192.168.1.36.49434	40.79.173.40.443	ESTABLISHED
tcp4	0	0	*.9999	*.*	LISTEN
tcp6	0	0	2806:268:3403:87.49432	2600:1901:1:7c5:.443	ESTABLISHED
tcp6	0	0	2806:268:3403:87.49418	2607:f8b0:4012:8.443	ESTABLISHED
tcp6	0	0	2806:268:3403:87.49417	2607:f8b0:4012:8.443	ESTABLISHED
tcp6	0	0	2806:268:3403:87.49415	2607:f8b0:4012:8.443	ESTABLISHED
tcp4	0	0	192.168.1.36.49402	194.124.205.65.443	ESTABLISHED
tcp4	0	0	192.168.1.36.49398	71.18.251.23.443	ESTABLISHED
tcp6	0	0	2806:268:3403:87.49389	2600:9000:26a5:4.443	ESTABLISHED
tcp4	0	0	192.168.1.36.49382	184.87.197.26.443	ESTABLISHED
tcp4	0	0	192.168.1.36.49381	184.87.197.26.443	ESTABLISHED
tcp4	0	0	192.168.1.36.49380	184.87.197.26.443	ESTABLISHED
tcp4	0	0	192.168.1.36.49379	184.87.197.26.443	ESTABLISHED
tcp4	0	0	192.168.1.36.49378	184.87.197.26.443	ESTABLISHED
tcp4	0	0	192.168.1.36.49377	184.87.197.9.443	ESTABLISHED
tcp4	0	0	192.168.1.36.49374	184.87.197.9.443	ESTABLISHED
tcp4	0	0	192.168.1.36.49372	184.87.197.26.443	ESTABLISHED
tcp4	0	0	192.168.1.36.49371	184.87.197.9.443	ESTABLISHED
tcp4	0	0	192.168.1.36.49370	184.87.197.9.443	ESTABLISHED
tcp4	0	0	192.168.1.36.49369	184.87.197.9.443	ESTABLISHED
tcp4	0	0	192.168.1.36.49368	104.18.35.224.443	ESTABLISHED
tcp4	0	0	192.168.1.36.49367	104.18.35.224.443	ESTABLISHED
tcp4	0	0	192.168.1.36.49366	184.87.197.9.443	ESTABLISHED
tcp4	0	0	192.168.1.36.49364	18.97.36.61.443	ESTABLISHED
tcp4	0	0	192.168.1.36.49363	184.87.197.9.443	ESTABLISHED
tcp6	0	0	2806:268:3403:87.49361	2607:f8b0:4012:8.443	ESTABLISHED
tcp4	0	0	192.168.1.36.49360	35.165.195.87.443	ESTABLISHED
tcp4	0	0	192.168.1.36.49359	18.160.124.22.443	ESTABLISHED
tcp4	0	0	192.168.1.36.49358	18.160.124.124.443	ESTABLISHED
tcp4	0	0	192.168.1.36.49357	199.232.157.140.443	ESTABLISHED
tcp6	0	0	2806:268:3403:87.49356	2607:f8b0:4012:8.443	ESTABLISHED
tcp6	0	0	2806:268:3403:87.49355	2606:4700::6811:.443	ESTABLISHED
tcp6	0	0	2806:268:3403:87.49354	2607:f8b0:4012:8.443	ESTABLISHED
tcp6	0	0	2806:268:3403:87.49353	2607:f8b0:4012:8.443	ESTABLISHED
tcp6	0	0	2806:268:3403:87.49352	2607:f8b0:4012:8.443	ESTABLISHED
tcp6	0	0	2806:268:3403:87.49351	2607:f8b0:4012:8.443	ESTABLISHED
tcp6	0	0	2806:268:3403:87.49350	2607:f8b0:4012:8.443	ESTABLISHED
tcp6	0	0	2806:268:3403:87.49349	2607:f8b0:4012:8.443	ESTABLISHED

EXPLORER

PracticaScript#992

tcpserver.py

Welcome

tcpserver.py

```
1 import socket
2 import threading
3
4 bind_ip = "0.0.0.0"
5 bind_port = 9999
6
7 server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
8 server.bind((bind_ip, bind_port))
9 server.listen(5)
10
11 print(f"[*] Listening on {bind_ip}:{bind_port}")
12
13 def handle_client(client_socket):
14     request = client_socket.recv(1024)
15     print(f"[*] Received: {request.decode('utf-8', errors='ignore')}")
16
17     client_socket.send(b"ACK!")
18     client_socket.close()
19
20 while True:
21     client, addr = server.accept()
22     print(f"[*] Accepted connection from: {addr[0]}:{addr[1]}")
23
24     client_handler = threading.Thread(target=handle_client, args=(client,))
25     client_handler.start()
```

CHAT

nadxiellyvazquez - zsh - 120x23

tcp4	0	0	192.168.1.36.49434	40.79.173.40.443	ESTABLISHE
tcp4	0	0	*.9999	.*	LISTEN
tcp6	0	0	2806:268:3403:87.49432	2600:1901:1:7c5:443	ESTABLISHE
tcp6	0	0	2806:268:3403:87.49418	2607:f8b0:4012:8.443	ESTABLISHE
tcp6	0	0	2806:268:3403:87.49417	2607:f8b0:4012:8.443	ESTABLISHE
tcp6	0	0	2806:268:3403:87.49415	2607:f8b0:4012:8.443	ESTABLISHE
tcp4	0	0	192.168.1.36.49402	194.124.205.65.443	ESTABLISHE
tcp4	0	0	192.168.1.36.49398	71.18.251.23.443	ESTABLISHE
tcp6	0	0	2806:268:3403:87.49389	2600:9000:26a5:4.443	ESTABLISHE
tcp4	0	0	192.168.1.36.49382	184.87.197.26.443	ESTABLISHE
tcp4	0	0	192.168.1.36.49381	184.87.197.26.443	ESTABLISHE
tcp4	0	0	192.168.1.36.49380	184.87.197.26.443	ESTABLISHE
tcp4	0	0	192.168.1.36.49379	184.87.197.26.443	ESTABLISHE
tcp4	0	0	192.168.1.36.49378	184.87.197.26.443	ESTABLISHE
tcp4	0	0	192.168.1.36.49377	184.87.197.9.443	ESTABLISHE
tcp4	0	0	192.168.1.36.49374	184.87.197.9.443	ESTABLISHE
tcp4	0	0	192.168.1.36.49372	184.87.197.26.443	ESTABLISHE
tcp4	0	0	192.168.1.36.49371	184.87.197.9.443	ESTABLISHE
tcp4	0	0	192.168.1.36.49370	184.87.197.9.443	ESTABLISHE
tcp4	0	0	192.168.1.36.49369	184.87.197.9.443	ESTABLISHE
tcp4	0	0	192.168.1.36.49368	104.18.35.224.443	ESTABLISHE
tcp4	0	0	192.168.1.36.49367	104.18.35.224.443	ESTABLISHE

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
/usr/local/bin/python3 "/Volumes/NO NAME/PracticaScript#992/tcpserver.py"
/usr/local/bin/python3 "/Volumes/NO NAME/PracticaScript#992/tcpserver.py"
nadxiellyvazquez@192 PracticaScript#992 % /usr/local/bin/python3 "/Volumes/NO NAME/PracticaScript#992/tcpserver.py"
[*] Listening on 0.0.0.0:9999
[]
```

OUTLINE

TIMELINE

```
[nadxiellyvazquez@192 PracticaScript#992 % cd /Volumes/NAD/PracticaScript#992/tcpclient.py
cd: not a directory: /Volumes/NAD/PracticaScript#992/tcpclient.py
[nadxiellyvazquez@192 PracticaScript#992 % /Volumes/NAD/PracticaScript#992/tcpclient.py
zsh: permission denied: /Volumes/NAD/PracticaScript#992/tcpclient.py
[nadxiellyvazquez@192 PracticaScript#992 % /usr/local/bin/python3 ~/Desktop/PracticaScript#992/tcpclient.py
[*] Intentando conectar a 127.0.0.1:9999...
[*] ¡Conexión exitosa!
[*] El servidor respondió: ACK!
nadxiellyvazquez@192 PracticaScript#992 % █
```

Capturing from Loopback: lo0



tcp.port == 9999									
No.	Time	Source	Destination	Protocol	Length	Info			
370	435.860572	127.0.0.1	127.0.0.1	TCP	68	49618 → 9999	[SYN]	Seq=0 Win=65535 Len=0 MSS=16344 WS=64 TSval=2074029014 TSecr=0 SACK_PE...	
371	435.860644	127.0.0.1	127.0.0.1	TCP	68	9999 → 49618	[SYN, ACK]	Seq=0 Ack=1 Win=65535 Len=0 MSS=16344 WS=64 TSval=1898248045 TSecr=...	
372	435.860659	127.0.0.1	127.0.0.1	TCP	56	49618 → 9999	[ACK]	Seq=1 Ack=1 Win=408320 Len=0 TSval=2074029014 TSecr=1898248045	
373	435.860671	127.0.0.1	127.0.0.1	TCP	56	[TCP Window Update] 9999 → 49618 [ACK] Seq=1 Ack=1 Win=408320 Len=0 TSval=1898248045 TSecr=...			
374	435.860713	127.0.0.1	127.0.0.1	TCP	82	49618 → 9999	[PSH, ACK]	Seq=1 Ack=1 Win=408320 Len=26 TSval=2074029014 TSecr=1898248045	
375	435.860728	127.0.0.1	127.0.0.1	TCP	56	9999 → 49618	[ACK]	Seq=1 Ack=27 Win=408320 Len=0 TSval=1898248045 TSecr=2074029014	
376	435.861498	127.0.0.1	127.0.0.1	TCP	60	9999 → 49618	[PSH, ACK]	Seq=1 Ack=27 Win=408320 Len=4 TSval=1898248045 TSecr=2074029014	
377	435.861518	127.0.0.1	127.0.0.1	TCP	56	9999 → 49618	[FIN, ACK]	Seq=5 Ack=27 Win=408320 Len=0 TSval=1898248046 TSecr=2074029014	
378	435.861521	127.0.0.1	127.0.0.1	TCP	56	49618 → 9999	[ACK]	Seq=27 Ack=5 Win=408320 Len=0 TSval=2074029015 TSecr=1898248045	
379	435.861532	127.0.0.1	127.0.0.1	TCP	56	49618 → 9999	[ACK]	Seq=27 Ack=6 Win=408320 Len=0 TSval=2074029015 TSecr=1898248046	
380	435.863078	127.0.0.1	127.0.0.1	TCP	56	49618 → 9999	[FIN, ACK]	Seq=27 Ack=6 Win=408320 Len=0 TSval=2074029016 TSecr=1898248046	
381	435.863103	127.0.0.1	127.0.0.1	TCP	56	9999 → 49618	[ACK]	Seq=6 Ack=28 Win=408320 Len=0 TSval=1898248047 TSecr=2074029016	

> Frame 370: Packet, 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on in
> Null/Loopback
> Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
> Transmission Control Protocol, Src Port: 49618, Dst Port: 9999, Seq: 0, Len: 0

```
0000 02 00 00 00 45 00 00 40 00 00 40 00 40 06 00 00  ....E..@..@..@..
0010 7f 00 00 01 7f 00 00 01 c1 d2 27 0f de a1 f9 73  ....s
0020 00 00 00 00 b0 02 ff ff fe 34 00 00 02 04 3f d8  ....4...?
0030 01 03 03 06 01 01 08 0a 7b 9f 2b d6 00 00 00 00  ....{.+....
0040 04 02 00 00
```

```
.....E..@..@..@..
.....s
.....4...?
.....{.+....
.....
```

wireshark_LoopbackRJC8L3.pcapng Packets: 389 · Displayed: 12 (3.1%) Profile: Default

OUTLINE
TIMELINE

tcpclient.py

Describe what to build

+ Auto

Local Default Approvals


```
nadxiellyvazquez@192 PracticaScript#992 % netstat -a -n -p TCP | grep 9999  
tcp4      0      0 *.9999      *.*          LISTEN  
nadxiellyvazquez@192 PracticaScript#992 %
```